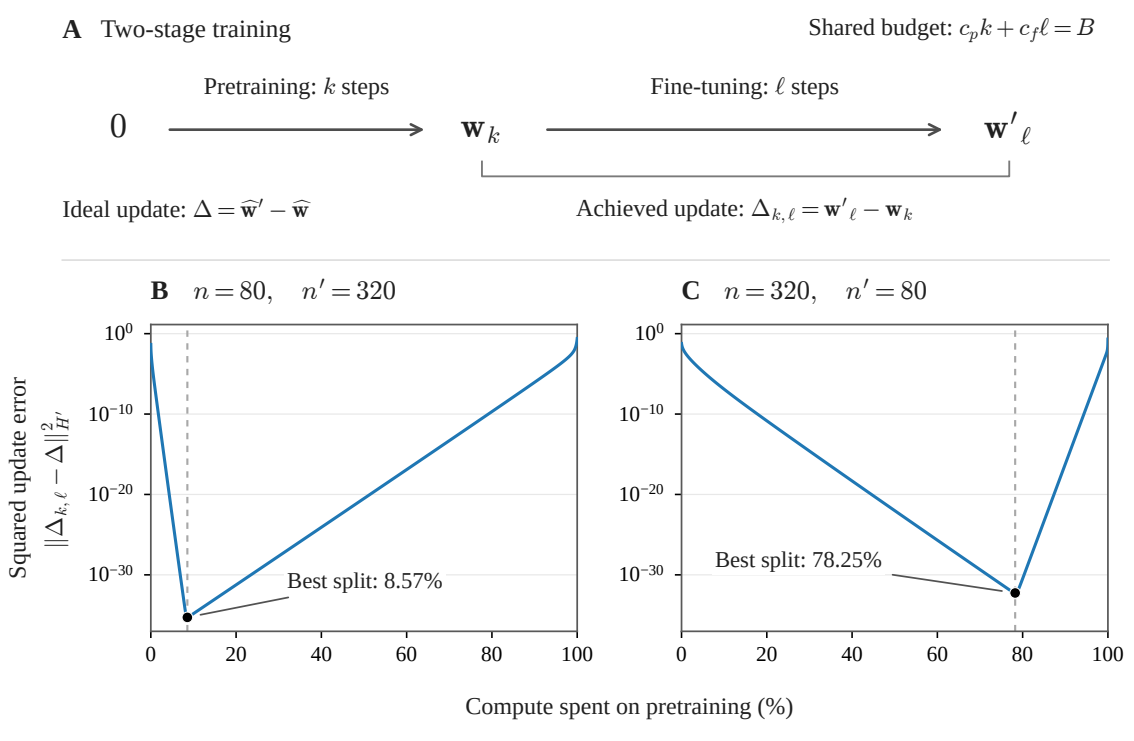

A Two-stage training
Shared budget: $c_p k + c_f \ell = B$
Pretraining: k steps
Fine-tuning: ℓ steps
0
$\mathbf{w}_k$
$\mathbf{w'}_\ell$
Ideal update: $\Delta = \widehat{\mathbf{w}}' - \widehat{\mathbf{w}}$
Achieved update: $\Delta_{k,\ell} = \mathbf{w'}_\ell - \mathbf{w}_k$
B $n = 80, \quad n' = 320$
C $n = 320, \quad n' = 80$
Squared update error
$\|\Delta_{k,\ell} - \Delta\|_{H'}^2$
10^0
10^{-10}
10^{-20}
10^{-30}
Best split: 8.57%
Best split: 78.25%
0
20
40
60
80
100
Compute spent on pretraining (%)